

答案

2026 年 7 月 5 日

1. 解：设曲面方程为 $F(x, y, z) = x^3 + y^2 + z^3 - 1 = 0$ ，求偏导得：

$$F_x = 3x^2, \quad F_y = 2y, \quad F_z = 3z^2$$

在点 $P(-1, 1, 1)$ 处的法向量为 $\vec{n} = (3, 2, 3)$. 切平面方程为：

$$3(x + 1) + 2(y - 1) + 3(z - 1) = 0 \implies 3x + 2y + 3z - 2 = 0$$

法线方程为：

$$\frac{x + 1}{3} = \frac{y - 1}{2} = \frac{z - 1}{3}$$

2. 解：函数 $f(x, y) = x - 2y$ 在区域 $D : (x - 1)^2 + (y + 2)^2 \leq 5$ 上. 内部无驻点，在边界上使用拉格朗日乘数法，设 $L = x - 2y + \lambda((x - 1)^2 + (y + 2)^2 - 5)$. 解得驻点为 $(0, 0)$ 和 $(2, -4)$. 经计算： $f(0, 0) = 0$, $f(2, -4) = 10$. 故最大值为 10，最小值为 0.

3. 解：

(a) $\int_0^2 dx \int_0^{2-x} (2x - y^2) dy = \int_0^2 (4x - 2x^2 - \frac{1}{3}(2-x)^3) dx = \frac{4}{3}$

(b) 极坐标变换： $\int_0^{2\pi} d\theta \int_0^1 \frac{r}{(1+r^2)^2} dr = 2\pi \cdot [-\frac{1}{2(1+r^2)}]_0^1 = \frac{\pi}{2}$

(c) 球坐标变换 ($\Omega : r \leq \cos \varphi$): $\int_0^{2\pi} d\theta \int_0^{\pi/2} \sin \varphi d\varphi \int_0^{\cos \varphi} r^3 dr = \frac{\pi}{10}$

4. 解：

(a) $\int_1^5 (x-1)^{-1/2} dx = [2\sqrt{x-1}]_1^5 = 4$

(b) 令 $t = \sqrt{x-2}$, 得 $I = \int_0^\infty \frac{2tdt}{(t^2+9)t} = 2 \cdot [\frac{1}{3} \arctan(\frac{t}{3})]_0^\infty = \frac{\pi}{3}$

5. 解：级数 $\sum_{n=0}^\infty (n+1)^2 (\frac{x}{3})^n$ 收敛域为 $(-3, 3)$. 由 $\sum_{n=0}^\infty u^{n+1} = \frac{u}{1-u}$ 求导得 $\sum_{n=0}^\infty (n+1)u^n = \frac{1}{(1-u)^2}$.

再次对 u 求导得 $\sum_{n=0}^\infty (n+1)^2 u^n = \frac{1}{(1-u)^2} + \frac{2u}{(1-u)^3} = \frac{1+u}{(1-u)^3}$. 代入 $u = x/3$ 得和函数：

$$S(x) = \frac{1+x/3}{(1-x/3)^3} = \frac{9(3+x)}{(3-x)^3}$$

6. 解：作偶延拓，计算傅里叶系数： $a_0 = \frac{2}{\pi} \int_0^\pi (\pi - x) dx = \frac{2}{\pi} [\pi x - \frac{1}{2} x^2]_0^\pi = \pi$. $a_n = \frac{2}{\pi} \int_0^\pi (\pi - x) \cos nx dx = \frac{2}{\pi} [\frac{\pi - x}{n} \sin nx]_0^\pi + \frac{2}{n\pi} \int_0^\pi \sin nx dx = \frac{2(1 - \cos n\pi)}{n^2 \pi}$. 当 $n = 2k - 1$ 时 $a_n = \frac{4}{(2k-1)^2 \pi}$, 故：

$$f(x) = \frac{\pi}{2} + \sum_{k=1}^\infty \frac{4}{\pi(2k-1)^2} \cos(2k-1)x$$

7. 解:

- (a) **曲线积分:** 积分函数 $P = \frac{-y}{x^2 + 4y^2}, Q = \frac{x}{x^2 + 4y^2}$. 在原点 $(0, 0)$ 处函数无定义, 但易验证 $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = \frac{4y^2 - x^2}{(x^2 + 4y^2)^2}$. 取辅助圆周 $L_1: x^2 + y^2 = r^2$ (足够小, 含于 L 内部, 取逆时针方向), 由格林公式的复连通区域性质:

$$\oint_L = \oint_{L_1} = \iint_{D'} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy + \oint_{L_1} = 0 + \oint_{L_1} \frac{x dy - y dx}{x^2 + 4y^2}$$

在 L_1 上, $x = r \cos \theta, y = r \sin \theta, dx = -r \sin \theta d\theta, dy = r \cos \theta d\theta$:

$$\oint_{L_1} = \int_0^{2\pi} \frac{r^2 \cos^2 \theta + r^2 \sin^2 \theta}{r^2 \cos^2 \theta + 4r^2 \sin^2 \theta} d\theta = \int_0^{2\pi} \frac{1}{\cos^2 \theta + 4 \sin^2 \theta} d\theta = 4 \int_0^{\pi/2} \frac{d\theta}{\cos^2 \theta + 4 \sin^2 \theta}$$

分子分母同时除以 $\cos^2 \theta$:

$$= 4 \int_0^{\pi/2} \frac{\sec^2 \theta}{1 + 4 \tan^2 \theta} d\theta = 4 \int_0^\infty \frac{d(\tan \theta)}{1 + (2 \tan \theta)^2} = 2 \int_0^\infty \frac{du}{1 + u^2} = 2 \cdot [\arctan u]_0^\infty = \pi$$

- (b) **曲面积分:** 投影区域 $D_{xy}: 2x + 2y \leq 1, x \geq 0, y \geq 0$. 曲面 $z = 1 - 2x - 2y$.

$$dS = \sqrt{1 + z_x^2 + z_y^2} dx dy = \sqrt{1 + (-2)^2 + (-2)^2} dx dy = 3 dx dy$$

代入被积函数: $8x + 8y + 4z = 8x + 8y + 4(1 - 2x - 2y) = 4$.

$$I = \iint_D 4 \cdot 3 dx dy = 12 \cdot \text{Area}(D) = 12 \cdot \left(\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \right) = \frac{3}{2}$$

8. 解:

设 Σ 为抛物面 $z = 3 - x^2 - y^2 (z > 0)$ 的上侧. 补底面 $\Sigma_1: z = 0$ (取下侧, 即法向量指向负 z 方向), 构成封闭曲面 $\Sigma + \Sigma_1$, 包围区域为 Ω .

由高斯公式:

$$\iint_{\Sigma} + \iint_{\Sigma_1} = \iiint_{\Omega} \text{div} \vec{F} dV$$

$$\text{其中 } \text{div} \vec{F} = \frac{\partial(yz)}{\partial x} + \frac{\partial(zx)}{\partial y} + \frac{\partial(z(x^2 + y^2))}{\partial z} = 0 + 0 + (x^2 + y^2) = x^2 + y^2.$$

- (a) **三重积分:** 采用柱坐标, $0 \leq \theta \leq 2\pi, 0 \leq r \leq \sqrt{3}, 0 \leq z \leq 3 - r^2$:

$$\iiint_{\Omega} r^2 \cdot r dr d\theta dz = 2\pi \int_0^{\sqrt{3}} r^3 (3 - r^2) dr = 2\pi \left[\frac{3}{4} r^4 - \frac{1}{6} r^6 \right]_0^{\sqrt{3}} = 2\pi \left(\frac{27}{4} - \frac{27}{6} \right) = \frac{9\pi}{2}$$

- (b) **补面 Σ_1 :** Σ_1 为 $z = 0$ 的圆盘. 因为在 Σ_1 上 $z = 0$, 代入第二类曲面积分公式:

$$\iint_{\Sigma_1} yz dy dz + zx dz dx + z(x^2 + y^2) dx dy = \iint_{\Sigma_1} 0 dy dz + 0 dz dx + 0 dx dy = 0$$

$$\text{故所求积分为 } I = \frac{9\pi}{2} - 0 = \frac{9\pi}{2}$$